

```
1  --@block
2  CREATE TABLE `Customer` (
3      `id` INT NOT NULL PRIMARY KEY,
4      `name` VARCHAR(50) NOT NULL,
5      `address` VARCHAR(50) NOT NULL,
6      `phone` BIGINT NULL,
7      `email` VARCHAR(50) NOT NULL,
8      `birthdate` DATE NOT NULL,
9      `age` BIGINT NULL
10 );
11 CREATE TABLE `Offering` (
12     `id` INT NOT NULL PRIMARY KEY,
13     `description` VARCHAR(100) NOT NULL
14 );
15 CREATE TABLE `Product` (
16     `offering_id` INT NOT NULL PRIMARY KEY,
17     FOREIGN KEY (`offering_id`) REFERENCES `Offering`(`id`),
18     `name` VARCHAR(50) NOT NULL,
19     `first_release_date` DATE NOT NULL,
20     `standard_price` DECIMAL(8, 2) NOT NULL
21 );
22 CREATE TABLE `Service` (
23     `id` INT NOT NULL,
24     `offering_id` INT NOT NULL,
25     FOREIGN KEY (`offering_id`) REFERENCES `Offering`(`id`),
26     PRIMARY KEY(`id`, `offering_id`),
27     `company_unit` VARCHAR(50) NOT NULL,
28     `service_condition` VARCHAR(100) NOT NULL
29 );
30 CREATE INDEX idx_service_id ON service(id);
```

```
31 CREATE TABLE `Repair`(  
32     `service_id` INT NOT NULL PRIMARY KEY,  
33     FOREIGN KEY (service_id) REFERENCES `Service`(id),  
34     `cost` DECIMAL(8, 2) NOT NULL  
35 );  
36 CREATE TABLE `Maintenance`(  
37     `service_id` INT NOT NULL PRIMARY KEY,  
38     FOREIGN KEY (service_id) REFERENCES `Service`(id),  
39     `hourly_rate` DECIMAL(4, 2) NOT NULL  
40 );  
41 CREATE TABLE `Purchase`(  
42     `offering_id` INT NOT NULL,  
43     FOREIGN KEY (offering_id) REFERENCES `Offering`(id),  
44     `customer_id` INT NOT NULL,  
45     FOREIGN KEY (customer_id) REFERENCES `Customer`(id),  
46     PRIMARY KEY (customer_id, offering_id),  
47     `date_purchased` DATE NOT NULL,  
48     `contact_person` VARCHAR(50) NOT NULL  
49 );  
50 CREATE TABLE `Bill`(  
51     `service_id` INT NOT NULL,  
52     FOREIGN KEY (service_id) REFERENCES `Service`(id),  
53     `customer_id` INT NOT NULL,  
54     FOREIGN KEY (customer_id) REFERENCES `Customer`(id),  
55     PRIMARY KEY(service_id, customer_id),  
56     `service_date` DATE NOT NULL,  
57     `amount_due` DECIMAL(8,2) NOT NULL,  
58     `date_due` DATE NOT NULL,  
59     `client_id` INT NULL  
60 );
```

```

62 --@DLOCK
63 INSERT INTO customer (id, name, address, phone, email, birthdate, age)
64 VALUES (1, 'Hanson Jekyll', 'Atlanta, GA', 4705456789, 'hodge@companyname.com', '1976-08-15', 49),
65 (2, 'Joanna Perl', 'Alpharetta, GA', 4047349991, 'jPerl@company-name.com', '1992-07-23', 33),
66 (3, 'Arel Castro', 'Sacramento, CA', 3419095522, 'areliC@company-name2.com', '1998-11-03', 27),
67 (4, 'Chase Dereks', 'Dallas, TX', 8066752342, 'chase6@companyname.net', '1994-05-30', 31),
68 (5, 'Julie Evans', 'Richmond, VA', 6863412390, 'julieE@companyname2.com', '1981-12-17', 43),
69 (6, 'John Russo', 'Brooklyn, NY', NULL, 'russoJ@co-name.com', '1998-05-27', 27),
70 (7, 'Sandy Martinez', 'Jonesboro, GA', NULL, 'sandyM@companyname2.net', '2000-04-04', 25),
71 (8, 'Emial James', 'Atlanta, GA', 4701238973, 'Ejames@gmail.com', '1990-02-23', 35),
72 (9, 'Name Phoenix', 'Phoenix, Arizona', 4808939291, 'Namephoenix@gmail.com', '1995-03-12', 30),
73 (10, 'Burna Boy', 'Lagos, Nigeria', 2341029945, 'burna@gmail.com', '1991-07-02', 34),
74 (11, 'Hikari Mitsushima', 'Brooklyn, New York', 2120981234, 'hikariM@gmail.com', '1985-11-30', 40),
75 (12, 'Jason Derulo', 'Miramar, Florida', 4219801821, 'derulo@gmail.com', '1989-11-21', 36),
76 (13, 'Lonnie Rashid Lynn', 'Chicago, Illinois', 2138930321, 'lonnierashid@gmail.com', '1972-03-13', 53)
77 ;
78 INSERT INTO offering(id, description)
79 VALUES (1, 'Product'),
80 (2, 'Product'),
81 (3, 'Service'),
82 (4, 'Product'),
83 (5, 'Service'),
84 (6, 'Service'),
85 (7, 'Service'),
86 (8, 'Product'),
87 (9, 'Service'),
88 (10, 'Service'),
89 (11, 'Service'),
90 (12, 'Service'),
91 (13, 'Service')
92 ;
93 INSERT INTO product(offering_id, name, first_release_date, standard_price)
94 VALUES (1, 'Laptop', '2000-02-02', 2599.00),
95 (2, 'Desktop Computer', '1990-07-07', 850.00),
96 (4, 'Solid State Drive', '2016-09-09', 90.00),
97 (8, 'Wi-Fi Router', '2003-04-04', 45.99)
98 ;
99 INSERT INTO service(offering_id, id, company_unit, service_condition)
100 VALUES (3, 1, 'Maintenance', 'Remote security patch installations and software updates for Company-Name2 lapt'),
101 (5, 2, 'Repair', 'Upgrade of RAM module for laptops at CompanyName.'),
102 (6, 3, 'Repair', 'Data recovery from 2 failed SSDs at Co-Name'),
103 (7, 4, 'Maintenance', 'Replace power adapter of 2 routers for CompanyName2.'),
104 (9, 5, 'Maintenance', 'Onsite firmware update to router at CompanyName2'),
105 (10, 6, 'Repair', 'Replace PSU of 4 desktops at Company-Name.'),
106 (11, 7, 'Maintenance', 'TRIM command optimization and scheduling for SSDs at Co-name.'),
107 (12, 8, 'Repair', 'Screen panel replacement of laptop at Company-Name.'),
108 (13, 9, 'Maintenance', 'Remote software update of laptops and desktops at Company-Name.')
109 ;

```

PROBLEMS 15

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

QUERY RESULTS

SQL CONSOLE

SQL CONSOLE

```

110 INSERT INTO Repair(service_id, cost)
111 VALUES (2, 40963.78),
112 (3, 4320.56),
113 (6, 105.88),
114 (8, 250.45)
115 ;
116 INSERT INTO Maintenance(service_id, hourly_rate)
117 VALUES (1, 92.50),
118 (4, 85.00),
119 (5, 95.00),
120 (7, 80.00),
121 (9, 92.50)
122 ;
123 INSERT INTO purchase(offering_id, customer_id, date_purchased, contact_person)
124 VALUES (1, 1, '2025-03-12', 'Lilly Smith'),
125 (2, 2, '2025-02-21', 'Jeffery Lewis'),
126 (3,3, '2025-10-09', 'John Thomas'),
127 (4, 4, '2025-07-18', 'Layla Jackson'),
128 (5, 5, '2024-12-26', 'Lebron James'),
129 (6,7, '2010-09-23', 'Black Panther'),
130 (7,6, '2012-08-12', 'Celine Dion'),
131 (8,8, '2025-10-09', 'Denzel Washington'),
132 (9,9, '2016-05-10', 'Starlan Treadwell'),
133 (10,10, '2018-04-23', 'Kaydence Sartorius'),
134 (11,11, '2024-11-07', 'Vlasta Burian'),
135 (12,12, '2023-10-18', 'Silvio Santos'),
136 (13,13, '2022-03-26', 'Lee Jae-Yong')
137 ;
138 INSERT INTO bill(service_id, customer_id, service_date, amount_due, date_due)
139 VALUES (1, 3, '2023-10-23', 277.50, '2023-11-24'),
140 (2,2, '2021-09-23', 40963.78, '2021-10-23'),
141 (4, 1, '2025-05-12', 170.00, '2025-06-01'),
142 (5, 5, '2024-03-01', 190.00, '2024-03-11'),
143 (7, 10, '2025-10-07', 240.00, '2025-10-21'),
144 (9, 11, '2025-06-23', 92.50, '2025-07-12'),
145 (3, 13, '2023-09-02', 4320.56, '2023-09-07'),
146 (6,7, '2022-03-07', 105.88, '2022-03-14'),
147 (8, 9, '2023-04-19', 250.45, '2023-04-21')
148 ;
149

```

154 D:\Run on active connection | P: Select block
155 --@lock
156 Select c.*, p.offering_id, p.date_purchased, p.contact_person
157 From customer c
158 Inner join purchase p
159 on c.id = p.customer_id
160 ;

id	name	address	phone	email	birthdate	age	offering_id	date_purchased	contact_person
1	Hanson Jekyll	Atlanta, GA	4705456789	hHodge@companyname.c...	1976-08-15	49	1	2025-03-12	Lilly Smith
2	Joanna Peri	Alpharetta, GA	4047349991	JPeri@company-name.com	1992-07-23	33	2	2025-02-21	Jeffery Lewis
3	Arel Castro	Sacramento, CA	3419095522	arelC@company-name2.co...	1998-11-03	27	3	2025-10-09	John Thomas
4	Chase Dereks	Dallas, TX	8066752342	chase6@companyname.net	1994-05-30	31	4	2025-07-18	Layla Jackson
5	Julie Evans	Richmond, VA	6863412390	JulieE@companyname2.com	1981-12-17	43	5	2024-12-26	Lebron James
6	John Russo	Brooklyn, NY	NULL	russol@co-name.com	1998-05-27	27	6	2012-08-12	Celine Dion
7	Sandy Martinez	Jonesboro, GA	NULL	sandyM@companyname2...	2000-04-04	25	7	2010-09-23	Black Panther
8	Emial James	Atlanta, GA	4701238973	EJames@gmail.com	1990-02-23	35	8	2025-10-09	Denzel Washington
9	Name Phoenix	Phoenix, Arizona	4808939291	Namephoenix@gmail.com	1995-03-12	30	9	2016-05-10	Starlan Treadwell
10	Burna Boy	Lagos, Nigeria	2341029945	burna@gmail.com	1991-07-02	34	10	2018-04-23	Kaydence Sartorius
11	Hikari Mitsushima	Brooklyn, New York	2120981234	hikariM@gmail.com	1985-11-30	40	11	2024-11-07	Vlasta Burian
12	Jason Derulo	Miramar, Florida	4219801821	derulo@gmail.com	1989-11-21	36	12	2023-10-18	Silvio Santos
13	Lonnie Rashid Lynn	Chicago, Illinois	2138930321	lonnierashid@gmail.com	1972-03-13	53	13	2022-03-26	Lee Jae-Yong

▷ Run on active connection | ≡ Select block

```
150 --@block
151 Select p.*
152 from product p
153 WHERE p.standard_price > 500
154 ;
155
```

techco_test2: Select p.* from... techco_test2: Select p.* from... X

offering_id	name	first_release_date	standard_price	
abc Filter...	abc Filter...	abc Filter...	abc Filter...	
1	Laptop	2000-02-02	2599.00	
2	Desktop Computer	1990-07-07	850.00	

▷ Run on active connection | ≡ Select block

```
--@block
UPDATE purchase
SET customer_id=8
WHERE offering_id=4;

UPDATE purchase
SET customer_id=9
WHERE offering_id=13;

UPDATE purchase
SET customer_id=9
WHERE offering_id=10;

UPDATE purchase
SET customer_id=9
WHERE offering_id=2;

UPDATE purchase
SET customer_id=12
WHERE offering_id=1;

UPDATE purchase
SET customer_id=3
WHERE offering_id=9;

UPDATE purchase
SET customer_id=6
WHERE offering_id=9;
```

210 Run on active connection | Select block

```

247 --@block
248 SELECT o.*, p.name, p.first_release_date, p.standard_price
249 FROM offering o
250 JOIN product p
251 WHERE o.description='Product' AND p.offering_id=o.id
252 ;

```

techco_test2: SELECT o.* FROM... techco_test2: SELECT o.*, p.na... X

id	description	name	first_release_date	standard_price
1	Product	Laptop	2000-02-02	2599.00
2	Product	Desktop Computer	1990-07-07	850.00
4	Product	Solid State Drive	2016-09-09	90.00
8	Product	Wi-Fi Router	2003-04-04	45.99

```

179 --@block
180 UPDATE bill
181 SET customer_id=11
182 WHERE customer_id=9;
183
184 UPDATE bill
185 SET customer_id=11
186 WHERE customer_id=7;
187
188 UPDATE bill
189 SET customer_id=13
190 WHERE customer_id=5;
191

```

212 Run on active connection | Select block

```

212 --@block Q.4
213 SELECT c.*, COUNT(b.customer_id) AS num_of_services
214 FROM customer c
215 JOIN bill b ON c.id=b.customer_id
216 GROUP BY c.id;

```

techco: SELECT c.*, COUNT... X

id	name	address	phone	email	birthdate	age	num_of_services
1	Hanson Jekyll	Atlanta, GA	4705456789	hHodge@companyname.c...	1976-08-15	49	1
2	Joanna Perl	Alpharetta, GA	4047349991	jPeri@company-name.com	1992-07-23	33	1
3	Arelli Castro	Sacramento, CA	3419095522	areliC@company-name2.co...	1998-11-03	27	1
10	Burna Boy	Lagos, Nigeria	2341029945	burna@gmail.com	1991-07-02	34	1
11	Hikari Mitsushima	Brooklyn, New York	2120981234	hikariM@gmail.com	1985-11-30	40	3
13	Lonnie Rashid Lynn	Chicago, Illinois	2138930321	lonnierashid@gmail.com	1972-03-13	53	2

```
192  --@block
193  UPDATE bill
194  SET client_id=2
195  WHERE service_id=8;
196
197  UPDATE bill
198  SET client_id=6
199  WHERE service_id=2;
200
201  UPDATE bill
202  SET client_id=19
203  WHERE service_id=5;
```

230 --@block Q.5

231 SELECT b.service_id, b.amount_due, c.name, s.company_unit, s.service_condition

232 FROM bill b

233 JOIN customer c ON c.id=b.customer_id

234 JOIN service s ON s.id=b.service_id

235 ORDER BY amount_due DESC

236 LIMIT 1

237 ;

238

techco: SELECT MIN(b.amo...

techco: SELECT * FROM bi...

techco: SELECT MAX(b.amo...

techco: SELECT b.amount_...

techco: SE

service_id	amount_due	name	company_unit	service_condition
Filter...	Filter...	Filter...	Filter...	Filter...
2	40963.78	Joanna Perl	Repair	Upgrade of RAM module for laptops at CompanyName.

239 --@block Q.6

240 SELECT b.service_date, b.customer_id, b.service_id, s.offering_id, s.company_unit, s.service_condition

241 FROM bill b

242 JOIN service s ON b.service_id=s.id

243 ORDER BY b.service_date DESC

244 LIMIT 1

245 ;

SELECT * FROM se...

techco: SELECT b.service...

techco: SELECT b.service...

techco: SELECT b.service...

techco: SELECT * FROM bi...

techco: SELECT * FROM se...

techco: SELE

service_date	customer_id	service_id	offering_id	company_unit	service_condition
Filter...	Filter...	Filter...	Filter...	Filter...	Filter...
2025-10-07	10	7	11	Maintenance	TRIM command optimization and scheduling for SSDs at Co-name.